

NextGen Empirical Finance Research Camp — Week 7 Lesson Plan

Independent Research Project I — From Question to Pre-Analysis Plan

Prof. Lei Gao · George Mason University · Costello College of Business Session:

Friday, August 7, 2026 · 12:00-2:00 pm ET (Zoom).

— Where this session sits

Weeks 1–6 built the toolkit: probability and inference, OLS and its geometry, causal inference, quasi-experiments, reading the empirical canon, and text/AI as data. The last two homeworks were about consuming research — the Week 5 **Paper Teardown** and the Week 6 **verified AI literature map**. This week is the pivot every student has been climbing toward: **from consuming research to producing it**.

Each student now has a topic and a literature map. Today we turn that into a **falsifiable, pre-registered empirical design with data in hand** — the design half of the capstone paper due **August 15**. The rule of the day: by 2:00 pm, every student walks out with a one-page design their paper can actually be built on. We are not writing results today; we are committing to a plan honest enough that a null result would still be informative.

This maps directly to **textbook Week 7** (Ch 7.1–7.5, Lab 7). The running anchor is Prof. Gao's *The Rainbow of Municipal Credit* — a staggered difference-in-differences design with a crisp identifying assumption and an explicit threats table, exactly the artifact each student will build for themselves.

— Learning objectives — by the end, a student can...

1. Apply the **"so-what" filter** and a **feasibility × novelty** check to lock a researchable question. 2. Write a one-sentence **identifying assumption** and a **threats-and-responses table** for their own design. 3. Draft a short-form **pre-analysis plan** (Olken 2015): directional hypotheses, one primary specification, multiple-testing control (BH-FDR), and falsification tests — registered **before** peeking at outcomes. 4. Pull data **reproducibly** (cache-or-fetch + a pinned vintage + an audit log) from the right source — WRDS (CRSP/Compustat), EDGAR, FRED, HMDA, 13F, PatentsView, or yfinance. 5. Build a clean analysis dataset that avoids **look-ahead, survivorship, and leakage** (identifier crosswalks, winsorizing, principled missing-data handling).

— Materials

- **Slides** — Week 7 GMU deck (24 slides, full speaker notes).
- **This lesson plan** and the **homework** — Pre-Analysis Plan + Identification Memo.
- **Textbook Week 7** — Ch 7.1 (Idea Generation), 7.2 (Data Acquisition), 7.3 (Pre-Analysis Plan), 7.4 (Building the Analysis Dataset), 7.5 (Identification Memo); **Lab 7** (your data, reproducibly); notebooks **nb7.1-nb7.5**.

same-sex-marriage-ban adoption → municipal borrowing costs. We use its **identifying assumption** and **threats table** as the model artifact. Secondary anchors: Gao & Sun (2019, PNAS) fair lending and Deng, Gao & Kim (2020, JCF) Reg SHO — both clean natural experiments.

— Timed agenda (120 minutes)

0:00-0:10 — Welcome, recap & collect the deliverable (10 min)

Collect the Week 6 **AI literature map + verification log**. Recap the arc: you have read the frontier; today you plant your own flag on it. Frame the session honestly — the goal is not a finished result but a **design you would defend to a skeptic**, filed before you see the answer.

0:10-0:45 — Block A: From a question to a design (35 min)

1. **The "so-what" filter.** Three questions kill most ideas in ninety seconds: Is the answer already known? Would anyone change a decision if they knew it? Can you actually measure both the cause and the effect? A question that survives all three is worth your two weeks. 2. **Feasibility × novelty (the 2×2).** Aim for the **feasible-and-fresh** quadrant. Avoid heroic-data (a perfect question you can't get data for) and been-done (clean data, no new question). 3. **From question to a design — the cast → dataset map.** Who/what/when in your question maps to which dataset. If you can't name the dataset, you don't yet have a project. 4. **The identifying-assumption sentence.** One sentence, filled in: "Comparing [treated units] to [controls], my estimate of [effect] is causal if [assumption holds], which is plausible because [design feature]." Worked live on Rainbow: ban vs. not-yet-ban states pay municipal spreads that would have moved in parallel absent the ban — plausible because adoption timing is driven by court/legislative calendars, not by credit trajectories. 5. **The threats-and-responses table.** For each threat — confounder, reverse causality, selection, measurement error — write the **design response**. This table is the referee's first page; building it now is how you referee yourself before anyone else does.

0:45-0:50 — Break (5 min)

0:50-1:25 — Block B: The pre-analysis plan, and getting the data (35 min)

1. **Why pre-register.** The garden of forking paths: with enough post-hoc choices, noise becomes "significance." Committing the primary specification **before** you see outcomes is what makes a null result publishable and an effect believable. 2. **The Olken short-form PAP.** Directional **hypotheses**; the **one** primary specification (outcome, treatment, sample, controls, fixed effects, clustering — all named); a **multiple-testing** correction (Benjamini-Hochberg FDR) if you test several outcomes; and **falsification / placebo** tests that should return nothing if your design is clean. 3. **Data acquisition in practice.** Pick the right source — CRSP/Compustat via WRDS, EDGAR filings, FRED macro, HMDA mortgages, 13F holdings, PatentsView, or yfinance — and pull it **reproducibly**: cache-or-fetch, **pin a data vintage**, keep an audit log, and never hand-edit a downloaded file. 4. **Building the analysis dataset.** Join on the right **identifier crosswalk** (PERMNO ↔ GVKEY ↔ CIK); guard against **survivorship** and **look-ahead** bias (use an as-of merge, not a naïve join); **winsorize** outliers by a stated rule; and handle missing data on purpose, not by silent dropping. 5. **The one rule.** File the PAP as a

tagged commit before the first outcome regression. The timestamp is the evidence you did not fish.

1:25-1:55 — Block C: Design workshop — lock your own project (30 min)

Each student drafts, **for their own project**, three things on one page: (a) the **identifying-assumption sentence**, (b) a **3-row threats-and-responses table**, and (c) the **primary-specification line** (outcome ~ treatment + controls + FE, clustered by ____). **Peer round-robin (2 min each)**: present your identifying assumption out loud; the room's job is to name **one threat you missed**. Catching it here is free; catching it in August is not.

1:55-2:00 — Wrap & assignment (5 min)

Assign the **Pre-Analysis Plan + Identification Memo** (with a reproducible **data-pull plan**) for each student's own project — **due before the Week 8 session (Friday, August 14)**. This is the **design half** of the August 15 paper; Week 8 is where you execute it.

— Instructor notes

- **Protect Block C.** The workshop is where a generic design becomes their design. Keep it moving; two crisp minutes per student beats one long critique.
- **Anchor everything to a real paper.** Put Rainbow's actual identifying assumption and threats table on the screen — students copy structure far better than they copy abstractions.
- **Tie back to the AI week.** A proposal like the cybersecurity-breach study (recall: match the estimator to the outcome horizon — an event study for the direct price reaction vs. staggered DiD for the slower peer/vendor contagion) is the perfect live example of "the design decision is the research."
- **Keep the PAP short.** One to two pages. The value is **commitment**, not length; resist scope creep. A tight PAP filed on time beats a sprawling one filed never.
- **Verified anchors only.** Use Prof. Gao's confirmed papers (Rainbow of Municipal Credit; Gao & Sun 2019, PNAS; Deng, Gao & Kim 2020, JCF) — model the citation discipline the whole camp teaches.